

Additionally, Nodes implement a separate 32-bit counter for each [Check-In Protocol](#) use case:

- *Check-In Counter*

Technical details for how each counter type works are described in the following sections. [Table 16](#), “[Message Counter Type Overview](#)” is provided to summarize higher-level differences between Message Counter Types:

Table 16. Message Counter Type Overview

Message Counter Type	Session Type	Lifetime	Rollover Policy	Nonvolatile
Global Unencrypted	Unsecured	Unlimited	Allowed	Optional
Global Encrypted Data	Group	Operational Group Key	Allowed	Mandatory
Global Encrypted Control	Group	Operational Group Key	Allowed	Mandatory
Secure Session	Unicast	Session Key	Expires	Optional
Check-In Counter	Unsecured	Unlimited	Allowed	Mandatory

4.6.1.1. Message Counter Initialization

All message counters SHALL be initialized with a random value using the `Crypto_DRBG(len = 28) + 1` primitive. Message counters are initialized to a random number to increase the difficulty of traffic analysis attacks by making it harder to determine how long a particular session has been open. The random initializer ranges from 1 to 2^{28} in order to maximize initial entropy while still reserving the vast majority of the range to actual counter values (roughly $2^{32} - 2^{28}$).

4.6.1.2. Global Unencrypted Message Counter

All Nodes SHALL implement an unencrypted message counter, which is used to generate counters for unencrypted messages.

Typically, Nodes store the *Global Unencrypted Message Counter* in RAM. This makes the counter subject to loss whenever the system reboots or otherwise loses its state. This is permissible because retaining the *Global Unencrypted Message Counter* is not essential to the confidentiality or integrity of the message. In the event that the *Global Unencrypted Message Counter* for a Node is lost, Nodes SHALL randomize the initial value of this counter on startup per [Section 4.6.1.1](#), “[Message Counter Initialization](#)”.

4.6.1.3. Global Group Encrypted Message Counters

The *Global Group Encrypted Message Counters* are used to generate the counter for messages encrypted using group keys. There are two such counters:

- The *Global Group Encrypted Data Message Counter* is used to encode regular data messages encrypted with a group key.
- The *Global Group Encrypted Control Message Counter* is used to encode [control messages](#) encrypted with a group key.